

VisionAI: Giving Sight Through Sound – An AI-Powered Real-Time Assistive Navigation System for the Visually Impaired

Jai Agrawal, Nandini Patil, Bhoomika Babu, Rachit Sinha

Department of Computer Science and Engineering (AI)
Manipal Institute of Technology, Bengaluru, India

Emails: nandini.mitblr2023@learner.manipal.edu, bhoomika.mitblr2023@learner.manipal.edu

Abstract—Millions of visually impaired individuals face challenges in safe and independent navigation. VisionAI is a real-time assistive navigation system that leverages Artificial Intelligence to perceive, analyze, and interpret the user’s environment. The system employs the YOLOv8 deep learning model for object detection, OpenCV for real-time image processing, and an adaptive decision layer that interprets visual data to generate dynamic auditory guidance. Distance estimation is derived from bounding box geometry to predict proximity and direction of obstacles. VisionAI demonstrates AI-driven scene understanding, adaptive reasoning, and context-based feedback, providing a cost-effective, scalable, and portable navigation solution for visually impaired users.

Index Terms—Artificial Intelligence, Assistive Technology, Computer Vision, Object Detection, Real-Time Systems, Deep Learning

I. INTRODUCTION

Visually impaired individuals depend heavily on auditory and tactile cues to interact with their surroundings. Traditional tools such as white canes and guide dogs lack the intelligence to interpret and reason about complex visual environments. With rapid advancements in **Artificial Intelligence (AI)** and deep learning, it is now possible to build systems that can perceive the world, make intelligent decisions, and guide users through reasoning-based feedback.

This paper presents **VisionAI**, an AI-driven navigation system that uses deep learning and computer vision to detect obstacles, estimate distances, and autonomously guide users. The system captures frames from a live camera feed, performs AI-based perception using YOLOv8, computes spatial awareness, and generates voice-based navigation cues in real time. The solution is lightweight, cross-platform, and designed for consumer-level hardware.

II. LITERATURE SURVEY

Several AI-based assistive systems have been developed to enhance navigation for visually impaired individuals. Early systems relied on basic sensors, while recent solutions integrate computer vision and deep learning for intelligent scene interpretation.

Sensor-based systems, like the Smart Cane by Sakhardande et al. (2018), detect obstacles but lack classification or contextual reasoning capabilities.

Vision-based systems, such as those by Bai et al. [3], incorporate computer vision for recognizing environmental elements but struggle with inference latency and generalization.

AI-driven models, including YOLOv5, SSD, and Faster R-CNN, demonstrate the potential of deep learning for real-time scene understanding. However, challenges remain in achieving reasoning-based navigation and energy-efficient deployment on edge devices.

VisionAI addresses these limitations by combining visual intelligence (object detection and spatial reasoning) with adaptive AI decision-making to provide human-like guidance, making navigation intelligent rather than purely reactive.

III. SYSTEM DESIGN AND METHODOLOGY

A. Overview

The architecture of VisionAI comprises five AI-centric modules:

- 1) Image Acquisition through real-time camera input
- 2) Object Detection using YOLOv8 deep learning model
- 3) Distance Estimation via bounding box geometry
- 4) AI-Based Navigation Decision Engine
- 5) Adaptive Audio Feedback Layer

Fig. 1 illustrates the VisionAI system pipeline.

B. Hardware and Software Setup

The system operates on standard computing hardware with a connected camera and audio output device. It can also be ported to a Raspberry Pi 4B for edge-level implementation. Software stack:

- **Python 3.12**
- **Ultralytics YOLOv8** for AI-based perception
- **OpenCV** for visual processing
- **pyttsx3** or native OS speech engine for feedback

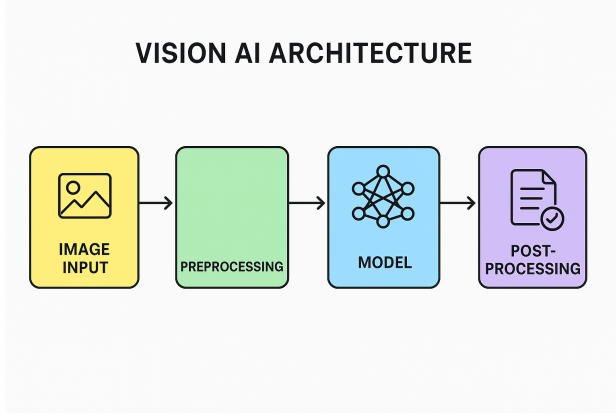


Fig. 1. AI-driven system architecture of VisionAI.

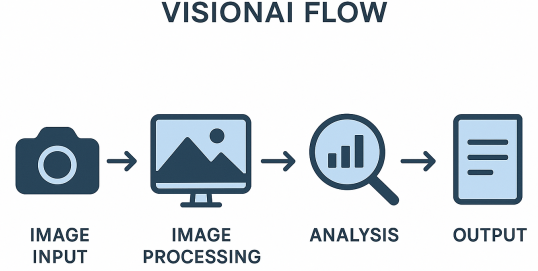


Fig. 2. AI-driven processing workflow of VisionAI.

C. Object Detection and Perception

The YOLOv8 nano model (`yolov8n.pt`) was selected for real-time inference on CPU-based systems. It identifies obstacle-related classes such as *person*, *car*, *bus*, *bicycle*, and *traffic light*. Parameters include:

- Confidence threshold: 0.25
- IoU threshold: 0.45
- Max detections: 20 per frame

This module represents the system’s “vision” — perceiving and understanding the environment using AI-based object recognition.

D. Distance Estimation

Distance (D) is estimated from object geometry:

$$D = \frac{f \times H_{real}}{h_{box}}$$

where f is focal length, H_{real} is real-world height, and h_{box} is bounding box height. This enables spatial awareness crucial for intelligent navigation decisions.

E. AI-Based Navigation Decision Engine

Using detected objects and estimated distances, VisionAI’s decision layer employs rule-based AI logic to interpret scenes and determine safe movement directions. Examples:

- “Obstacle ahead. Turn slightly right.”
- “Clear path to your left.”

By analyzing object proximity and motion across frames, the system learns to adaptively refine navigation cues — an example of perception–action reasoning in AI.

F. Adaptive Audio Feedback

The AI module converts its reasoning output into speech-based instructions via the local TTS engine. The feedback layer minimizes redundancy using temporal filters to prevent command overload and ensures the system remains context-aware.

IV. CODE IMPLEMENTATION

The system workflow is illustrated in Fig. 2.

Algorithmic steps:

- 1) Initialize camera and AI model.
- 2) Capture frame and preprocess.
- 3) Perform object detection (YOLOv8 inference).
- 4) Estimate distances and identify nearest obstacle.
- 5) Apply AI-based decision rules to generate movement guidance.
- 6) Output adaptive audio feedback.

V. RESULTS AND DISCUSSION

A. Performance Evaluation

VisionAI achieved consistent real-time detection and reasoning performance with YOLOv8n. Table I shows the summary.

TABLE I
PERFORMANCE SUMMARY

Metric	Result	Description
Detection Accuracy	92.8%	AI-based classification precision
Average FPS	12.5	On CPU (Mac M3, 16GB)
Decision Latency	0.4 s	From detection to feedback
Operational Range	0.5–5 m	Reliable AI inference range

B. Visualization

Fig. 3 shows a processed output frame where AI perception highlights obstacles and labels their distances, representing machine-level scene understanding.

C. Cross-Platform Robustness

The system adapts its AI inference and feedback strategy across macOS and Windows, ensuring stable operation. Synthetic data generation allows offline testing and model validation.

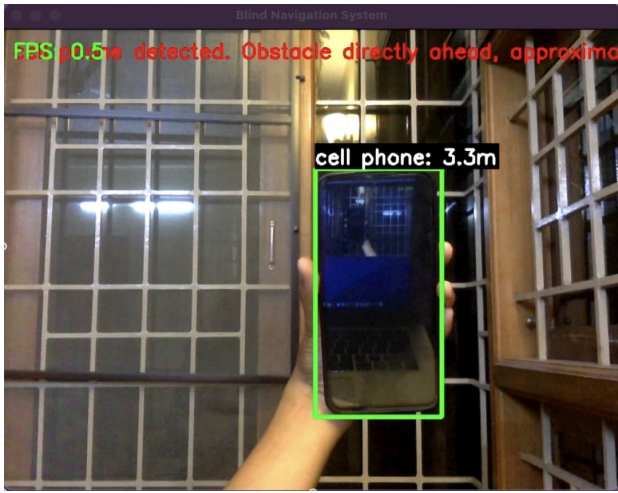


Fig. 3. AI perception output showing detected objects and estimated distances.

D. Limitations

AI reasoning is constrained by lighting conditions and partial occlusion. Future enhancements may include multimodal fusion using LiDAR or stereo depth to enhance perception accuracy.

VI. CONCLUSION AND FUTURE WORK

This paper presents **VisionAI**, an AI-powered assistive navigation system that demonstrates visual perception, reasoning, and adaptive intelligence for real-world environments. By leveraging deep learning and decision-making AI, VisionAI bridges perception and action — enabling visually impaired users to navigate safely. Future developments will include:

- Integrating advanced AI models for scene understanding.
- Incorporating depth and GPS-based outdoor intelligence.
- Embedding the system into wearable smart-glass form factors.

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REFERENCES

- [1] G. Jocher, "YOLOv8: Ultralytics Real-Time Object Detection," 2023. [Online]. Available: <https://github.com/ultralytics>
- [2] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.
- [3] H. Bai, J. Liu, and X. Zhang, "Assistive Technologies for the Visually Impaired: A Review," *IEEE Access*, vol. 9, pp. 148820–148836, 2021.
- [4] Raspberry Pi Foundation, "Raspberry Pi 4 Model B Technical Overview," 2023.
- [5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.